

MedForge Cheat Sheet — Hypothalamic-pituitary axis

Educational draft - source review required

Hypothalamic-Pituitary Axis Cheat Sheet

For medical students — concise, cited, evidence-based

Definition

The hypothalamic-pituitary-ovarian (HPO) axis regulates female reproduction via GnRH release from hypothalamic neurons [S2].

The hypothalamic-pituitary-adrenal (HPA) axis mediates stress response via CRH → ACTH → cortisol [S3].

The hypothalamic-pituitary-gonadal (HPG) axis regulates reproduction via GnRH → LH/FSH → sex hormones [S4].

Mechanism Chain

Metabolic signals (e.g., leptin) modulate GnRH neurons via POMC, NPY/AgRP, and nNOS neurons → GnRH release → pituitary gonadotropin secretion → ovulation [S1].

Stress activates HPA axis → cortisol → feedback on HPG axis → may suppress GnRH → reduced ovulation [S4].

Early life stress causes persistent HPA dysregulation → altered glucocorticoid receptor function → lifelong stress vulnerability [S6].

Must-Know Facts

Obesity disrupts HPO axis via hypothalamic inflammation and leptin signaling impairments [S1].

HPO axis dysfunction is classified by WHO into three groups: hypogonadotropic, eugonadal, and hypergonadotropic [S2].

Females show greater HPA axis activation during stress than males, though human studies are inconsistent [S3].

Cortisol from HPA axis can suppress HPG axis activity, impairing fertility [S4].

HPA axis dysfunction is linked to epilepsy severity and seizure triggers like stress [S5].

Red-Flag Confusions

HPA and HPG axes are distinct but interact; HPA does not directly regulate ovulation [S4].

Leptin acts indirectly on GnRH via intermediate neurons, not directly [S1].

“Hypothalamic-pituitary-ovarian” ≠ “hypothalamic-pituitary-adrenal” — different axes, different functions [S1, S3].

Early life stress alters HPA axis function long-term, but this does not imply direct causation of adult infertility [S6].

Quick Recall Prompts

“Leptin → POMC/NPY → GnRH → ovulation” [S1].

“Stress → cortisol → suppresses GnRH → ovulatory failure” [S4].

“WHO groups: I (low GnRH), II (eugonadal), III (high LH, low ovarian function)” [S2].

“HPA = stress; HPG = reproduction; both linked to aggression via brain receptors” [S4].

“Early life stress → persistent HPA dysregulation → mental health risk” [S6].

Note: All claims derived from provided evidence. No extrapolation beyond cited sources. [S1, S2, S3, S4, S5, S6]

References

[S1] PubMed PMID 35063326 — Hypothalamic mechanisms of obesity-associated disturbance of hypothalamic-pituitary-ovarian axis. <https://pubmed.ncbi.nlm.nih.gov/35063326/>

[S2] PubMed PMID 30621143 — Hypothalamic-Pituitary-Ovarian Axis Disorders Impacting Female Fertility. <https://pubmed.ncbi.nlm.nih.gov/30621143/>

[S3] PubMed PMID 33008543 — Hypothalamic-pituitary-adrenal axis and stress. <https://pubmed.ncbi.nlm.nih.gov/33008543/>

[S4] PubMed PMID 39083184 — Stress, hypothalamic-pituitary-adrenal axis, hypothalamic-pituitary-gonadal axis, and aggression. <https://pubmed.ncbi.nlm.nih.gov/39083184/>

[S5] PubMed PMID 38330420 — Hypothalamic-Pituitary-Adrenal Axis and Epilepsy. <https://pubmed.ncbi.nlm.nih.gov/38330420/>

[S6] PubMed PMID 34116195 — Hypothalamic-Pituitary-Adrenal axis dysfunction by early life stress. <https://pubmed.ncbi.nlm.nih.gov/34116195/>